STATISTICAL COMPUTING

MINI REPORT

Random Variables, Transformations, Moments and Probability Distributions

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Subject	Statistical Computing
Report Type	Mini Report

Scope: This report focuses on Random Variables and the subsequent topics shown in the supplied outline. Conditional Probability and Bayes' Theorem are intentionally excluded.

1. Introduction

Probability provides a mathematical language for uncertainty, while random variables and probability distributions convert uncertain outcomes into quantities that can be analysed, predicted and computed. In statistical computing and Big Data Analytics, these concepts are used to model observations, estimate risk, forecast future events, detect unusual behaviour and support data-driven decisions.

The central idea is simple: a random variable represents an uncertain quantity; its distribution describes how likely its possible values are; transformations create useful derived variables; and moments summarise important properties such as location, variability, asymmetry and tail behaviour.

A useful practical pipeline is: Data → Random Variable → Distribution → Parameters → Prediction/Inference → Decision.

2. Random Variables

Definition

A random variable (RV) is a numerical function that assigns a real number to each outcome of a random experiment. It allows uncertain outcomes to be represented mathematically.

Notation: X denotes a random variable and x denotes one of its possible values.

Example

Suppose a data centre records the number of server requests arriving in one minute. Let X = number of requests in the next minute. X is a random variable because the exact number is unknown before the minute begins.

What does it enable us to do?

- Convert real-world uncertainty into a mathematical quantity that algorithms can process.
- Estimate expected demand, risk and variability rather than relying only on individual observations.
- Build probabilistic models for prediction and simulation.

Real-life applications and actions

- Cloud computing: predict incoming requests and decide how many servers/resources to allocate.
- Finance: model daily return or loss and quantify portfolio risk.
- Healthcare: represent patient outcomes or waiting times for planning and risk analysis.
- E-commerce: model number of purchases or customer activity to support inventory and recommendation systems.
- Cybersecurity: model login attempts or unusual network events to identify anomalous behaviour.

Big Data / Statistical Computing connection

At scale, millions of observations can be treated as realizations of one or more random variables. Distributed systems can then estimate their empirical distributions, means, variances and other statistics.

3. Discrete Random Variables

Definition

A discrete random variable takes a finite or countably infinite set of distinct values, usually integers. Its probability model is described by a probability mass function (PMF):

$$P(X = x) = p(x), \quad \text{with} \quad p(x) \geq 0 \quad \text{and} \quad \Sigma\, p(x) = 1$$

Example

Let X be the number of defective products in a sample of 10 products. Possible values are 0, 1, 2, …, 10, so X is discrete.

Real-life uses

- Number of customers arriving at a service desk in an hour.
- Number of failed transactions in a payment system.
- Number of clicks, purchases or messages generated during a time interval.
- Number of disease cases reported in a region during a period.

What scientists/analysts do with it

1. Estimate the probability of a particular count or range of counts.
2. Predict whether demand, failures or events will cross a critical threshold.
3. Choose capacity, staffing or inventory levels using calculated probabilities.
4. Simulate future event counts when direct experimentation is expensive or impossible.

4. Continuous Random Variables

Definition

A continuous random variable can take any value within an interval of real numbers. Its behaviour is described by a probability density function (PDF), f(x). For a continuous RV, the probability of one exact point is zero; probabilities are obtained over intervals:

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

Example

Let X be the response time of a web request, measured in milliseconds. X can take values such as 101.2 ms, 101.25 ms, 101.251 ms, etc., so it is naturally modelled as continuous.

Real-life uses

- Network latency and response time analysis.
- Measurement of temperature, pressure, weight and sensor signals.
- Time-to-failure of machines and electronic components.
- Patient recovery time or duration of hospital stay.
- Financial returns and measurement errors.

Prediction and action

An analyst may calculate P(X > 200 ms) to estimate the probability of a slow response. If that probability becomes too large, engineers can add servers, optimise code or change the system architecture. Thus a probability calculation can directly support an operational decision.

5. Transformations of Random Variables

Definition

A transformation of a random variable creates a new random variable from an existing one. If X is a random variable and Y = g(X), then Y is a transformed random variable.

$$\mathbf{Y = g(X)}$$

Simple example

Suppose X is temperature in Celsius. If a model requires Fahrenheit, define Y = 1.8X + 32. The new variable Y is a transformation of X.

Why transform data?

- Change the scale or units of measurement.
- Reduce skewness and make a distribution easier to model.
- Create meaningful business/scientific features from raw measurements.
- Make a model satisfy assumptions more closely.

Important transformations in practice

Transformation	Typical purpose	Example use
Log: Y = log(X)	Reduce right-skew / compress large values	Income, transaction amounts, population
Square root: Y = √X	Variance stabilisation in some count data	Counts and event frequencies
Standardisation: Z = (X−μ)/σ	Put variables on a common scale	Machine learning features
Linear: Y = aX+b	Change units or scale	Celsius → Fahrenheit

Advanced note: transformation of a continuous variable

For a one-to-one transformation Y = g(X), the PDF can be obtained using the change-of-variables principle. In one dimension, $f_Y(y) = f_X(x)\,|dx/dy|$, where $x = g^{-1}(y)$. This idea is fundamental in statistical modelling, simulation and probabilistic machine learning.

6. Moments of a Random Variable

Definition

Moments are numerical summaries of a probability distribution. They help describe its location, spread, asymmetry and tail behaviour.

Key moments

Moment / quantity	Meaning	Practical interpretation
1st raw moment: E[X]	Mean / expected value	Typical or long-run average
2nd central moment: Var(X)	Variance	Degree of variability / uncertainty
Standard deviation: √Var(X)	Spread in original units	Typical magnitude of deviation
3rd standardized moment	Skewness	Asymmetry of the distribution

4th standardized moment	Kurtosis	Tail/extreme-value behaviour

Example

If a server receives an average of 500 requests per minute but has very high variance, the average alone is not enough for capacity planning. The variance or standard deviation reveals how unstable demand is and helps determine a safer capacity level.

Real-life uses

- Risk management: measure average return and variability of investments.
- Machine learning: standardise features and diagnose distributional behaviour.
- Reliability engineering: study failure-time variability and extreme behaviour.
- Operations: estimate demand variability for staffing, inventory and resource allocation.

Moment Generating Function (MGF)

The moment generating function of X, when it exists around t = 0, is defined as:

$$M_X(t) = E[e^{tX}]$$

Moments can be obtained by differentiating the MGF at t = 0: $E[X^n] = M_X^{(n)}(0)$. MGFs are especially useful for deriving moments and identifying or studying probability distributions.

7. Important Probability Distributions

A probability distribution describes how probability is allocated over the possible values of a random variable. Choosing an appropriate distribution is a modelling decision: the goal is to represent the real data-generating process reasonably well.

Distribution	Type	Typical situation	Practical use
Bernoulli	Discrete	One trial: success/failure	Fraud/no fraud, click/no click
Binomial	Discrete	Number of successes in n trials	Defects, conversions, successes
Poisson	Discrete	Number of events in a fixed interval	Calls, arrivals, failures
Normal	Continuous	Approximately symmetric measurement variation	Measurement errors, aggregated quantities
Exponential	Continuous	Waiting time between events	Time to next arrival/failure
Uniform	Continuous	All values in an interval equally likely	Simulation and simple uncertainty models

Example: Poisson model

Suppose a server receives an average of 20 requests per second. A Poisson model can be used as a first model for the number of requests in a one-second interval. The analyst can estimate the probability of unusually high traffic and decide whether additional capacity or load balancing is needed.

Example: Normal model

If a manufacturing process produces a measurement that is approximately normally distributed, the mean and standard deviation can be used to estimate the proportion of items outside acceptable limits. This supports quality-control decisions.

8. Integrated Real-World Example: Big Data System Monitoring

Consider a large e-commerce platform monitoring its servers. The concepts in this report can be combined into one statistical workflow:

Concept	How it is used
Random variable	X = number of requests arriving in one minute.
Discrete RV	X is a count, so it takes integer values.
Distribution	A Poisson-like model may be considered for event counts, subject to model validation.
Continuous RV	Y = response time in milliseconds is continuous.
Transformation	Z = log(Y) may be used if response times are strongly right-skewed.
Moments	Mean and standard deviation summarise typical response time and variability.
Prediction	Estimate probabilities such as P(Y > service-level threshold).
Action	Scale resources, investigate anomalies or change system configuration when risk becomes high.

9. Statistical Computing Perspective

In modern data science, these concepts are rarely used only by hand. Statistical computing libraries allow analysts to estimate parameters, evaluate probability functions, simulate random variables and visualise distributions.

- Python: NumPy for numerical computation; SciPy for probability distributions and statistical functions; pandas for data manipulation; Matplotlib for visualisation.
- R: widely used for statistical modelling, probability distributions, simulation and research analysis.
- Big Data platforms: the same statistical ideas can be applied to large datasets using distributed processing frameworks when the data cannot be handled efficiently on a single machine.

A good research practice is to compare the theoretical distribution with the empirical distribution observed in data, validate assumptions, quantify uncertainty and report the limitations of the model.

1. Real-life example of the complete flow

Let's take a **hospital emergency department** as an example.

Suppose a hospital wants to answer:

"How many patients should we expect tomorrow, and how many doctors/nurses should we schedule?"

The complete statistical workflow is:

Data → Random Variable → Distribution → Parameters → Prediction/Inference → Decision

Step 1 — Data

The hospital collects the number of emergency patients arriving each day.

For example:

Day	Patients
Monday	82
Tuesday	91
Wednesday	87
Thursday	95
Friday	104
Saturday	118
Sunday	109

This is **observed data**.

At this stage, we are simply collecting measurements.

Step 2 — Random Variable

Now we define:

X = number of emergency patients arriving at the hospital in one day.

Why is X a **random variable**?

Because before tomorrow happens, we don't know the exact number.

Tomorrow could have:

- 85 patients
- 100 patients
- 120 patients
- 150 patients

The actual number is uncertain.

So:

\[X = \text{number of patients arriving in one day} \]

is a **discrete random variable**, because patients are counted in whole numbers.

Step 3 — Distribution

Now we ask:

How is X distributed?

Looking at historical data, suppose the hospital finds that daily arrivals behave approximately like a **Poisson distribution**.

Conceptually:

$$ X \sim Poisson(\lambda) $$

The distribution tells us **how probability is spread across possible patient counts**.

For example, instead of merely saying:

"Usually around 100 patients arrive."

we can ask much more useful questions:

What is the probability that more than 120 patients arrive tomorrow?

or

What is the probability that fewer than 80 patients arrive?

This is where probability distributions become powerful.

Step 4 — Parameters

A distribution has parameters that describe its behaviour.

For a Poisson distribution, the main parameter is:

$$ \lambda = \text{average number of events in the interval} $$

Suppose historical data gives:

$$ \lambda = 100 $$

So we have:

$$ X \sim Poisson(100) $$

Now the model has been **quantified**.

The parameter tells us the typical level of demand.

For other distributions, parameters may describe different properties.

For example, for a Normal distribution:

$$ X \sim N(\mu,\sigma^2) $$

where:

- μ = mean/location
- σ^2 = variance/spread

Step 5 — Prediction / Inference

Now comes the interesting part.

The hospital can use the model to answer questions such as:

What is the probability that tomorrow more than 120 patients will arrive?

Suppose the statistical calculation gives:

$$P(X>120)=0.022$$

That means approximately a **2.2% probability** of exceeding 120 patients under the assumed model.

But perhaps the hospital has a more important threshold:

What is the probability that demand exceeds the capacity of the current emergency department?

Suppose current capacity is 110 patients/day.

Then we calculate:

$$P(X>110)$$

This is **prediction**.

What about inference?

Inference goes in the other direction.

We have observed data and want to learn something about the underlying population/process.

For example:

"Based on the last 2 years of hospital records, what is the estimated average daily arrival rate?"

We might estimate:

$$\hat{\lambda}=103$$

and perhaps construct a confidence interval around that estimate.

So:

Prediction: What might happen tomorrow?

Inference: What can we learn about the underlying process from the data we already observed?

Step 6 — Decision

This is where statistics becomes practically valuable.

Suppose the hospital discovers:

$$P(X>110)=0.18$$

So there is an estimated **18% chance that daily emergency arrivals exceed current capacity**.

The hospital might decide to:

- schedule an additional doctor during peak periods,
- keep additional nurses on standby,
- increase available beds,
- prepare additional emergency equipment,
- create an overflow area,
- or increase staffing on particular days.

So the complete chain becomes:

Patient arrival records

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Random variable: X = daily patient arrivals

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Distribution: Poisson model

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Parameter: $\lambda \approx 100–103$ patients/day

↓

Prediction: Probability of exceeding capacity

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Decision: Increase staffing/resources when risk is sufficiently high

This is the key idea:

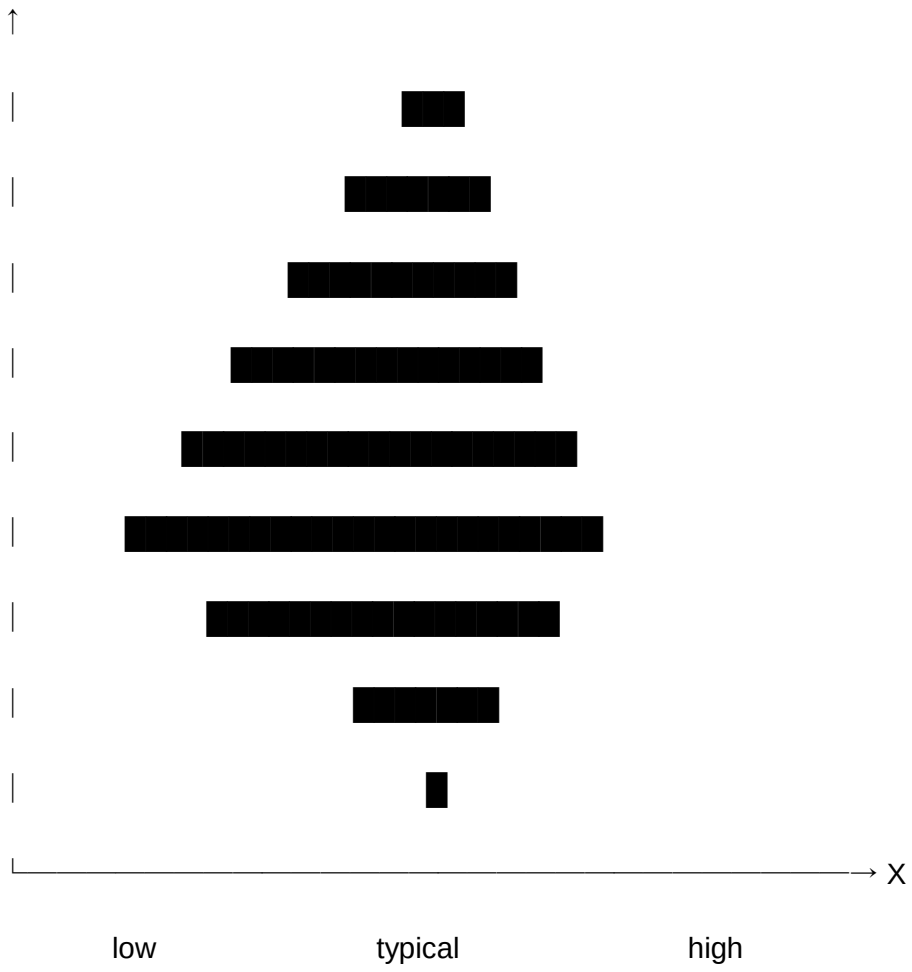
Statistics converts uncertainty into actionable decisions.

That is exactly why these concepts matter in **Big Data Analytics**.

2. Now, what does "tail behaviour" mean?

This is a very important concept, especially when you move toward **research, machine learning, finance, anomaly detection and risk analysis**.

Imagine a probability distribution plotted like this:



The **middle** represents the values that occur relatively commonly.

The **tails** are the extreme ends, So **tail behaviour** means:

How likely are extreme or unusually large/small values to occur?

3. A simple example

Imagine people's daily online spending.

Most people might spend:

₹500
₹800
₹1,200
₹1,500
₹2,000

But occasionally someone spends:

₹50,000
₹1,00,000
₹5,00,000

Those extremely large values are in the **right tail** of the distribution.

The important question becomes:

How much probability exists in that extreme region?

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4. Why does tail behaviour matter?

Because **rare events can have enormous consequences**.

Consider financial markets.

A model that understands only the average behaviour but ignores the tails can dramatically underestimate risk.

Example: Cybersecurity

Suppose a company's server normally receives:

10,000–20,000 requests per minute.

But suddenly:

500,000 requests/minute

arrive.

That's an extreme event.

A cybersecurity system is interested precisely in these **tail events**, because they might indicate:

- DDoS attacks
- bot activity
- abnormal traffic

- system abuse

So tail analysis can help answer:

"How unusual is this observation?"

and consequently:

"Should we trigger an alert?"

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5. Tail behaviour and the 4th moment: Kurtosis

This connects directly to your **Moments** topic.

The fourth standardized moment is related to **kurtosis**.

In simple terms:

Kurtosis tells us something about the tendency of a distribution to produce extreme observations / tail heaviness.

For example:

Thin/light tails

Extreme observations are relatively unlikely.

Heavy/fat tails

Extreme observations are considerably more likely.

This distinction is extremely important in:

- Finance → extreme market losses
- Insurance → catastrophic claims
- Cybersecurity → attack spikes
- Telecommunications → traffic bursts
- Engineering → rare failures
- Climate science → extreme weather
- Big Data → anomaly detection

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6. The most important distinction

Don't think:

Tail = simply the far end of the graph.

Think:

Tail behaviour = what happens probabilistically in the extreme regions of a distribution.

The scientist asks:

"How much probability is sitting in the extreme region?"

because that determines the likelihood of rare but potentially important events.

One excellent example for your presentation

You can say:

"Consider hospital emergency arrivals. Most days may have around 100 patients, but occasionally the hospital may receive 180 or 200 patients. These unusually high arrival counts belong to the right tail of the distribution. Studying the tail allows hospital administrators to estimate the probability of capacity-exceeding events and prepare additional staff and resources."

That is a much stronger explanation than simply saying **"tail means extreme values."**

10. Conclusion

Random variables provide the mathematical representation of uncertain quantities. Discrete and continuous random variables distinguish count-type and measurement-type uncertainty, while transformations allow data to be expressed in more useful forms. Moments summarise important characteristics of distributions, and probability distributions provide models for describing and predicting uncertain behaviour.

Together, these concepts form a practical foundation for statistical computing, Big Data Analytics and machine learning. Their real value is not only in calculating probabilities but in converting uncertainty into evidence-based actions—such as predicting demand, estimating risk, detecting abnormal behaviour, planning resources and improving system reliability.

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